

11. Create a DataFrame of ten rows and four columns with random values. Convert some values to NaN values. Highlight the NaN values.

Aim

To create a DataFrame with random values, replace some values with NaN, and highlight the NaN values.

Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame with random values.
3. Replace some values with NaN.
4. Highlight the NaN values.
5. Display the DataFrame.

Program and output

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randint(1,100,size=(10,4)),
                  columns=["A","B","C","D"])

df.iloc[2,1] = np.nan
df.iloc[5,3] = np.nan
df.iloc[7,0] = np.nan

def highlight_nan(value):
    if pd.isna(value):
        return "background-color: yellow"
    return ""

print(df)

styled_df = df.style.map(highlight_nan)

styled_df
```

	A	B	C	D
0	67.0	3.0	3	49.0
1	88.0	38.0	5	20.0
2	16.0	NaN	49	98.0
3	56.0	29.0	84	47.0
4	56.0	52.0	22	74.0
5	75.0	17.0	20	NaN
6	55.0	17.0	7	84.0
7	NaN	5.0	22	24.0
8	8.0	90.0	89	35.0
9	61.0	70.0	20	20.0

	A	B	C	D
0	67.000000	3.000000	3	49.000000
1	88.000000	38.000000	5	20.000000
2	16.000000	nan	49	98.000000
3	56.000000	29.000000	84	47.000000
4	56.000000	52.000000	22	74.000000
5	75.000000	17.000000	20	nan
6	55.000000	17.000000	7	84.000000
7	nan	5.000000	22	24.000000
8	8.000000	90.000000	89	35.000000
9	61.000000	70.000000	20	20.000000

Result

Thus, the DataFrame with NaN values was created and the NaN values were highlighted successfully.

12. Create a DataFrame of ten rows and four columns with random values. Set DataFrame background color to black and font color to yellow.

Aim

To create a DataFrame with random values and set the background color to black and font color to yellow.

Procedure

1. Import Pandas and NumPy.
2. Create a DataFrame with random values.
3. Apply black background color.
4. Apply yellow font color.
5. Display the styled DataFrame.

Program and output

```
 import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randint(1,100,size=(10,4)),
                  columns=["A","B","C","D"])

print(df)

styled_df = df.style.set_properties(**{
    'background-color': 'black',
    'color': 'yellow'
})

styled_df
```

```
...   A  B  C  D
0  49 73 81 50
1  28 53 32 79
2  25 20 90 44
3  93 64 17 42
4  84 11 73 80
5  52 90  6 92
6   9 26 52  4
7  31 92 28 30
8   5 21 99 90
9  59 62 75 84
```

	A	B	C	D
0	49	73	81	50
1	28	53	32	79
2	25	20	90	44
3	93	64	17	42
4	84	11	73	80
5	52	90	6	92
6	9	26	52	4
7	31	92	28	30
8	5	21	99	90
9	59	62	75	84

Result

Thus, the DataFrame was displayed successfully with a **black background** and **yellow font color**.